

# Billy Chen

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## Profile

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UBC Engineering Physics undergraduate combining expertise in SolidWorks CAD, C/C++/Python, embedded systems, and rapid prototyping with multi-year international and supervisory leadership. Experienced in designing robotics and Mars rover mechanisms, with a strong track record of delivering multidisciplinary engineering solutions from concept to fabrication.

## Engineering Projects & Technical Experience

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### UBC Rover Design Team — Chassis Subteam Member

Sep 2025 – Present

- **Rocker-Bogie Joint Adapters:** Modeled suspension adapters in SolidWorks using iterative CAD design and physical stress testing to maximize joint articulation.
- **Castable Urethane Wheels:** Researched rubber hardness, calculated load and flex parameters, and casted urethane wheels; validated via Mini Rover testing before scaling to full-size rover design.
- **Deployable Communication Relay:** Designed and fabricated an autonomous deployable communications relay system to extend teleoperation range and signal stability during field trials.
- Engineered a rack-and-pinion deployment system and retractable tripod base, integrating custom electronics, Raspberry Pi servo control, and a claw arm interface to deploy from the main rover.
- **CIRC 2026:** Executed mission route planning and FMEA risk analysis for CIRC 2026, engineering a contingency servo-driven wire cutter that saved a primary task failure.

### Spider Robot Capstone Project

Sep 2021 – Present

- Designed, constructed, and programmed a multi-legged autonomous spider robot using embedded microcontrollers and a RaspberryPi
- Integrated ultrasonic sensors with onboard control logic to implement autonomous room navigation and real-time obstacle avoidance.
- Programmed a Python computer vision script using real-time camera feeds to detect red visual targets and translate coordinate tracking into inverse kinematics for dynamic robot motion.

## Leadership & Experience

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### Elections Canada — Elections Supervisor

Sep 2021 – Apr 2025

- Managed end-to-end voting site operations, supply chain logistics, and regulatory compliance for municipal and federal elections, leading temporary staff to execute post-election ballot auditing and real-time equipment troubleshooting under high-stakes conditions.

### International Volunteer & Language Instructor

Nov 2022 – Dec 2024

- Completed two years of full-time missionary service for The Church of Jesus Christ of Latter-day Saints in Cambodia, acquiring working fluency in Khmer while managing daily operations, volunteer training, public outreach, and community education programs.

## Technical Skills

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**Programming & Software:** Python, C, C++, Java, MATLAB, BSL, Data Analysis

**Design & Prototyping:** Onshape, SolidWorks CAD, 3D Printing, Resin Casting, Parametric Modeling, Water Jet Cutting, Circuit Analysis, Soldering, Arduino, RaspberryPi

**Languages:** English (Native), Mandarin (Fluent), Cambodian / Khmer (Professional Working Proficiency)

## Education

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### University of British Columbia

Sep 2025 – Present (Expected Jun 2030)

*Bachelor of Applied Science (BASc) – Engineering Physics | GPA: 4.33*